17 There are some small cubes and some large cubes in a bag.
The cubes are red or the cubes are yellow.

The ratio of the number of small cubes to the number of large cubes is 4 : 7

The ratio of the number of red cubes to the number of yellow cubes is 3 : 5

(a) Explain why the least possible number of cubes in the bag is 88

4 + 7 = 11 parts; 3 + 5 = 8 parts.

Least possible total = LCM(11, 8) = 88.

1 mark
LCM of 11 and 8 is 88.

(1)

All the small cubes are yellow.

(b) Work out the least possible number of large yellow cubes in the bag.

Scale 4 : 7 by 8 → small 32, large 56.

1 mark
Scale the 11-part ratio.

Scale 3 : 5 by 11 → red 33, yellow 55.

1 mark
Scale the 8-part ratio.

All 32 small cubes are yellow.

Large yellow = 55 − 32 = 23.

1 mark
23 large yellow cubes.

23

(3)

(Total for Question 17 is 4 marks)